1.7 Order of Operations

Think about the situation

Valli and her four friends went to a butter milk shop. Each had a cup of butter milk and paid ₹30, assuming that the cost of one cup of butter milk to be ₹6. But the shop keeper told that the cost of butter milk had increased by ₹2. Then, Valli decided to give ₹2 more and paid ₹32. But the shop keeper claimed that she had to pay ₹40. Who is correct?

Valli calculated as,	Shop keeper calculated as,
= (5 × 6) + 2	= 5 × (6 + 2)
= 30 + 2	= 5 × 8
= 32	= 40



The amount ₹40 claimed by the Shop keeper is correct. This confusion can be avoided by using the brackets in the correct places like 5 × (6 + 2).

The rule of order of operations is called as BIDMAS. In BIDMAS the operations done from left to right.

Expansion of BIDMAS	
B	Bracket ( )
I	Indices (you will learn it later)
D	Division ÷ or /
M	Multiplication ×
A	Addition +
S	Subtraction −

Now, we try to solve 9 + 5 × 2 by using BIDMAS,

9 + (5 × 2) = 9 + 10

= 19

Example 1.10

Simplify : 24 + 2 × 8 ÷ 2 − 1

Solution

24 + 2 × 8 ÷ 2 − 1	(given question)
= 24 + 2 × 4 − 1	(÷ operation, completed first)
= 24 + 8 − 1	(× operation, completed second)
= 32 − 1	(+ operation, completed third)
= 31	(− operation , completed last)

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